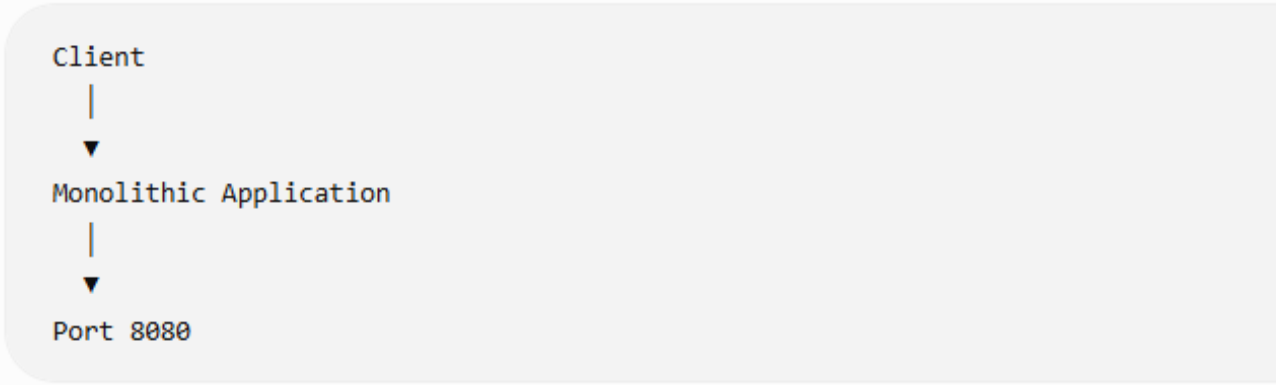
Structuring the Microservices And Port Numbers

1. Why Do We Need Different Ports?

Previously, we had a single monolithic application.

For example:



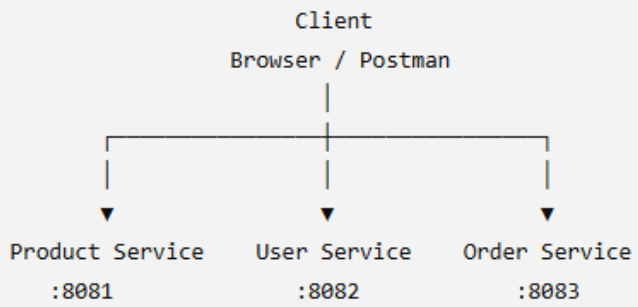
Now we are splitting the application into **three separate microservices**.

Since they are separate applications/processes, each one needs its own port.

2. Port Assignment

Microservice	Port
Product Microservice	`8081`
User Microservice	`8082`
Order Microservice	`8083`

So the architecture becomes:



3. Why can’t all three use 8080?

Because the three services will run independently on the **same machine** during local development.

A particular host port can normally be bound by only one listening application at a time.

So this would not work:

```
Product Service → 8080 ❌
User Service    → 8080 ❌
Order Service   → 8080 ❌
```

Instead:

```
Product Service → 8081 ✅
User Service    → 8082 ✅
Order Service   → 8083 ✅
```

Then we can access them independently:

```
Product → http://localhost:8081
User    → http://localhost:8082
Order   → http://localhost:8083
```

4. Remember

```
Monolith
   ↓
One application
   ↓
One port (e.g., 8080)


Microservices
   ↓
Multiple applications
   ↓
Multiple ports

Product → 8081
User    → 8082
Order   → 8083
```